

space	probe	per-ev.	$f^*(2\sigma)$, median over draws			SparKer range
			Maha.	MMD	SparKer	
SupCon	0.930	0.01	0.057	0.020	0.018	0.015–0.021
→ res (residual)	0.923	0.23	0.023	0.033	0.022	0.012–0.039
→ res (concat)	0.957	0.12	0.024	0.018	0.019	0.016–0.021
→ res-nplm (residual)	0.926	0.22	0.023	0.029	0.017	0.015–0.019
→ res-nplm (concat)	0.946	0.10	0.025	0.020	0.019	0.016–0.021
SupCon+SIGReg ($\lambda=5$)	0.912	0.02	$> 0.1^4$	0.031	0.037	0.032–0.042
→ res (residual)	0.902	0.01	$> 0.1^3$	0.044	0.069	0.044–0.091
→ res (concat)	0.949	0.01	$> 0.1^3$	0.035	0.042	0.027–0.060
→ res-nplm (residual)	0.921	0.16	0.044	0.038	0.022	0.017–0.025
→ res-nplm (concat)	0.943	0.06	$> 0.1^2$	0.033	0.030	0.020–0.036
NPLM-dist. (sup.)	0.788	0.17	0.038	0.039	0.027	0.024–0.040
→ res (residual)	0.884	0.03	$> 0.1^2$	0.046	0.046	0.044–0.077
→ res (concat)	0.908	0.03	$> 0.1^2$	0.042	0.043	0.033–0.059
→ res-nplm (residual)	0.841	0.01	$> 0.1^2$	0.044	0.039	0.024–0.049
→ res-nplm (concat)	0.879	0.09	0.047 ¹	0.045	0.031	0.018–0.043
NPLM-dist.+classwise SIGReg	0.938	0.21	0.032	0.033	0.027	0.024–0.029
→ res (residual)	0.917	0.03	$> 0.1^2$	0.037	0.048	0.039–0.084
→ res (concat)	0.958	0.03	0.034 ¹	0.035	0.047	0.037–0.078
→ res-nplm (residual)	0.948	0.07	0.034	0.036	0.043	0.028–0.079
→ res-nplm (concat)	0.966	0.17	0.023	0.035	0.037	0.018–0.046

Table 1: Pre-discovery: minimum injected fraction for a 2σ dataset-level detection, with and without the residual construction. f^* is the smallest fraction of the held-out class injected into a 5000-point corpus at which the median expected significance of the test reaches 2σ ; lower is better. The probe (oracle) and per-event power are shown for reference; the last column is SparKer’s range across draws. *Coverage*: CIFAR-10, leakage-free lineage, single holdout; draws 4, 7, 8, 9; 20/20 spaces on every draw; 200 null / 50 signal toys per fraction, $N_D=5000$; median over draws, superscript = draws where the test never reached 2σ by $f=0.1$.

Sigma-currency tables (exps 146/148), CIFAR-10 leakage-free lineage — generated September 6, 2026

$f^*(2\sigma)$ = smallest injected fraction at which the median expected significance of a dataset-level test reaches 2σ (200 null / 50 signal toys, $N_D=5000$); lower is better. Post tables: bold = best test per arm; † = the derived cut declined at some fraction below the crossing (the space is the frozen one there); purity ‘–’ = declined at $f=0.03$; ‘(n/6)’ = arm still running.

arm	draw	frozen SparKer	dist/95th	dist/derived	np/derived
SupCon	h4	0.015	0.027 (MMD)	0.014 (MMD) [†]	0.014 (MMD) [†]
	h7	0.021	0.035 (SparKer (anch.))	0.033 (SparKer (anch.))	0.018 (SparKer)
	h8	0.018	0.016 (Eucl. (anch.))	0.017 (MMD)	0.011 (Eucl. (anch.))
	h9	0.018	0.008 (Eucl.)	0.011 (Eucl. (anch.))	0.009 (SparKer)
SupCon+SIGReg ($\lambda=5$)	h4	0.042	0.021 (Eucl. (anch.))	0.028 (MMD)	0.028 (Eucl.)
	h7	0.034	0.021 (Eucl. (anch.))	0.036 (SparKer) [†]	0.036 (SparKer) [†]
	h8	0.040	0.015 (MMD)	0.021 (MMD)	0.020 (Eucl. (anch.))
	h9	0.032	0.016 (Eucl. (anch.))	0.026 (MMD) [†]	0.026 (MMD) [†]
NPLM-dist. (sup.)	h4	0.040	0.024 (Eucl. (anch.))	0.037 (Eucl. (anch.))	0.033 (Eucl. (anch.))
	h7	0.029	0.022 (Eucl. (anch.))	0.039 (Eucl. (anch.))	0.021 (Eucl. (anch.))
	h8	0.024	0.016 (SparKer (anch.))	0.015 (Eucl. (anch.))	0.016 (Eucl. (anch.))
	h9	0.024	0.009 (SparKer (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
NPLM-dist.+classwise SIGReg	h4	0.025	0.019 (Eucl. (anch.))	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))
	h7	0.029	0.018 (Eucl. (anch.))	0.027 (SparKer (anch.))	0.011 (Eucl. (anch.))
	h8	0.024	0.015 (Eucl. (anch.))	0.012 (Eucl. (anch.))	0.009 (Eucl. (anch.))
	h9	0.028	0.016 (Maha.)	0.020 (Eucl. (anch.))	0.020 (Eucl. (anch.))

Table 2: Does the discovery loop beat the frozen space? Frozen-space SparKer $f^*(2\sigma)$ against the best post-discovery test (named in parentheses) under three loop variants: the campaign’s distance pool with the 95th-percentile cut, the distance pool with the derived cut, and the density-ratio pool with the derived cut (the paper’s construction). Bold = the loop wins on that draw. *Coverage:* CIFAR-10, four parents x draws 4, 7, 8, 9; 48 complete (arm, draw, variant) cells, the loop beats the frozen space in 38; [†] = the derived cut had declined at the crossing fraction, so that number is the frozen space’s and is not counted as a win; ‘-’ = variant not run or still running.

dataset / backbone	best frozen (space / test, median f^*)		best discovery (arm / test)		gain
CIFAR-10 (scratch, 4 draws)	SupCon→res-nplm (resid.) / SparKer	0.017	SupCon / Eucl. (anch.)	0.013	1.3×
Galaxy10 / lejepa	SupCon / MMD	0.023	SupCon+SIGReg / Eucl. (anch.)	0.016	1.4×
Galaxy10 / visreg	SupCon→res (concat) / MMD	0.026	SupCon+SIGReg / Eucl. (anch.)	0.017	1.5×
Galaxy10 / dino	SupCon+SIGReg / MMD	0.029	SupCon / Eucl. (anch.)	0.019	1.5×
transductive ref. (lejepa)	gcd-ft / Maha.	0.013	–	–	sees the class
transductive ref. (visreg)	gcd-ft / Maha.	0.015	–	–	sees the class
transductive ref. (dino)	gcd-ft / MMD	0.028	–	–	sees the class

Table 3: The chosen algorithm, per dataset: best frozen space vs. best discovery pipeline in the same currency. Discovery wins everywhere, always through an anchor-aware test; the transductive GCD rows are the ceiling given the whole novel class unlabelled. *Coverage:* Medians over draws (censored cells imputed at 0.15); CIFAR-10 discovery = np pool + derived cut ($n_{\min}=10$), Galaxy10 discovery = np pool + derived cut ($n_{\min}=5$), both skip-on-refuse; frozen candidates exclude the transductive GCD arms (shown separately).

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	0.00	0.063	> 0.1	0.072	0.027	0.036	0.029
→ res (residual)	0.039	0.04	0.074	0.026	0.039	0.043	0.047	0.026
→ res-nplm (residual)	0.016	0.08	0.082	0.029	0.065	0.045	0.037	0.031
SupCon+SIGReg ($\lambda=5$)	0.042	0.01	0.033	0.021	> 0.1	0.027	0.031	0.039
→ res (residual)	0.091	0.01	0.068	0.043	0.035	0.036	0.044	0.045
→ res-nplm (residual)	0.023	0.10	0.027	0.017	0.077	0.028	0.029	0.037
NPLM-dist. (sup.)	0.040	0.10	> 0.1	0.024	0.087	0.046	0.046	0.043
→ res (residual)	0.077	0.01	> 0.1	> 0.1	0.081	0.035	0.043	0.041
→ res-nplm (residual)	0.049	0.01	> 0.1	0.047	0.087	0.062	0.083	0.044
NPLM-dist.+classwise SIGReg	0.025	0.20	0.034	0.019	0.033	0.044	0.028	0.024
→ res (residual)	0.084	0.01	0.027	0.027	0.033	0.033	0.029	0.027
→ res-nplm (residual)	0.079	0.03	0.043	0.022	0.037	0.036	0.047	0.041

Table 4: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, 95th-pct. cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 †	0.017†	–
→ res (residual)	0.039	0.03	0.042	0.039	0.068	0.039	0.040	0.032
→ res-nplm (residual)	0.016	0.01	0.047	0.039	0.049	0.034	0.027	0.024
SupCon+SIGReg ($\lambda=5$)	0.042	0.00	0.038	0.062	> 0.1	0.028	0.047	0.044
→ res (residual)	0.091	0.00	> 0.1	> 0.1	0.078	0.038	0.049	0.087
→ res-nplm (residual)	0.023	0.07	0.072	0.024	> 0.1	0.039	0.040	0.026
NPLM-dist. (sup.)	0.040	–	0.077	0.037	0.072	0.045	0.043	0.043
→ res (residual)	0.077	0.02	> 0.1	> 0.1	> 0.1	0.038	0.052	0.060
→ res-nplm (residual)	0.049	–	0.089	0.065	0.089	0.080	0.049 †	0.074
NPLM-dist.+classwise SIGReg	0.025	–	0.028†	0.020	0.074	0.029†	0.028†	0.021
→ res (residual)	0.084	–	> 0.1	0.050	0.083	0.041	0.047	0.050
→ res-nplm (residual)	0.079	–	0.049	0.050	0.046	0.043	0.067	0.050

Table 5: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 4, distance pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.015	–	> 0.1	–	> 0.1	0.014 [†]	0.017 [†]	–
→ res (residual)	0.039	0.21	0.033	0.015	0.024	0.041	0.033	0.029
→ res-nplm (residual)	0.016	0.14	0.035	0.020	0.039	0.041	0.037	0.034
SupCon+SIGReg ($\lambda=5$)	0.042	0.13	0.028	0.030	0.083	0.034	0.042	0.030
→ res (residual)	0.091	0.15	0.082	0.030	0.033	0.042	0.060	0.047
→ res-nplm (residual)	0.023	0.12	0.044	0.018	0.074	0.037	0.041	0.021
NPLM-dist. (sup.)	0.040	–	0.087	0.033	0.084	0.049	0.073	0.042
→ res (residual)	0.077	0.43	0.056	0.020	0.029	0.046	0.046	0.034
→ res-nplm (residual)	0.049	–	0.089	0.046	0.089	0.080	0.049 [†]	0.062
NPLM-dist.+classwise SIGReg	0.025	–	0.028 [†]	0.020	0.072	0.029 [†]	0.027 [†]	0.036
→ res (residual)	0.084	–	0.079	0.050	0.046	0.044	0.045	0.050
→ res-nplm (residual)	0.079	–	0.044	0.050	0.046	0.043	0.057	0.050

Table 6: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage*: CIFAR-10 holdout 4, density-ratio pool, derived cut; 12 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; [†] = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘–’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	> 0.1	> 0.1	0.043	0.047	0.052	0.035
SupCon+SIGReg ($\lambda=5$)	0.034	0.03	0.042	0.021	> 0.1	0.056	0.040	0.043
NPLM-dist. (sup.)	0.029	0.08	0.082	0.022	0.038	0.044	0.029	0.023
NPLM-dist.+classwise SIGReg	0.029	0.17	0.029	0.018	0.024	0.037	0.028	0.025

Table 7: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.00	0.071	> 0.1	0.036	0.056	0.040	0.033
SupCon+SIGReg ($\lambda=5$)	0.034	–	0.079	–	0.090	0.079	0.036 †	–
NPLM-dist. (sup.)	0.029	0.05	> 0.1	0.039	0.044	0.042	0.042	0.041
NPLM-dist.+classwise SIGReg	0.029	0.10	0.042	0.029	0.035	0.052	0.040	0.027

Table 8: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.021	0.68	0.022	0.020	0.021	0.052	0.018	0.020
SupCon+SIGReg ($\lambda=5$)	0.034	–	0.079	–	0.090	0.072	0.036 †	–
NPLM-dist. (sup.)	0.029	0.18	0.044	0.021	0.033	0.043	0.041	0.029
NPLM-dist.+classwise SIGReg	0.029	0.67	0.043	0.011	0.018	0.049	0.033	0.019

Table 9: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 7, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.039	0.016	0.023	0.018	0.021	0.017
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.026	0.019	> 0.1	0.015	0.026	0.023
NPLM-dist. (sup.)	0.024	0.16	0.037	0.017	0.019	0.019	0.021	0.016
NPLM-dist.+classwise SIGReg	0.024	0.18	0.017	0.015	0.022	0.022	0.017	0.015

Table 10: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.052	> 0.1	0.038	0.017	0.031	0.027
SupCon+SIGReg ($\lambda=5$)	0.040	0.01	0.024	0.076	0.062	0.021	0.029	0.029
NPLM-dist. (sup.)	0.024	0.59	0.063	0.015	0.025	0.023	0.023	0.017
NPLM-dist.+classwise SIGReg	0.024	0.55	0.042	0.012	0.039	0.036	0.035	0.021

Table 11: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.45	0.014	0.011	0.015	0.019	0.017	0.014
SupCon+SIGReg ($\lambda=5$)	0.040	0.22	0.024	0.020	0.024	0.025	0.032	0.035
NPLM-dist. (sup.)	0.024	0.64	0.046	0.016	0.041	0.024	0.018	0.017
NPLM-dist.+classwise SIGReg	0.024	0.73	0.075	0.009	0.037	0.044	0.040	0.021

Table 12: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 8, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.02	0.008	0.010	0.017	0.018	0.010	0.009
SupCon+SIGReg ($\lambda=5$)	0.032	0.01	0.025	0.016	0.040	0.040	0.025	0.025
NPLM-dist. (sup.)	0.024	0.14	0.017	0.014	0.015	0.018	0.014	0.009
NPLM-dist.+classwise SIGReg	0.028	0.17	0.016	0.022	0.016	0.029	0.017	0.017

Table 13: Post-discovery $f^*(2\sigma)$ on the distance pool, 95th-pct. cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, 95th-pct. cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.00	0.013	0.011	0.025	0.020	0.014	0.018
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	> 0.1	0.048	0.026 †	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.82	0.036	0.020	0.024	0.029	0.021	0.027
NPLM-dist.+classwise SIGReg	0.028	0.51	0.026	0.020	0.028	0.036	0.026	0.020

Table 14: Post-discovery $f^*(2\sigma)$ on the distance pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, distance pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

arm	frozen SparKer	purity	Eucl.	Eucl. (anch.)	Maha.	MMD	SparKer	SparKer (anch.)
SupCon	0.018	0.28	0.010	0.013	0.016	0.025	0.009	0.010
SupCon+SIGReg ($\lambda=5$)	0.032	–	0.047	0.050	0.045	0.026 †	0.027†	0.050
NPLM-dist. (sup.)	0.024	0.84	0.027	0.020	0.023	0.029	0.023	0.020
NPLM-dist.+classwise SIGReg	0.028	0.86	0.044	0.020	0.025	0.038	0.025	0.020

Table 15: Post-discovery $f^*(2\sigma)$ on the density-ratio pool, derived cut. The frozen-space SparKer f^* is repeated for reference; bold marks the best post-discovery test per arm. *Eucl. (anch.)* is the mean of $d_{\text{seen}} - d_{\text{disc}}$ and *SparKer (anch.)* seeds the kernels at the discovered anchors; both consume the anchors discovery planted, the other four do not. *Coverage:* CIFAR-10 holdout 9, density-ratio pool, derived cut; 4 arms; each fraction scored in its own fine-tuned space (2 rounds); 200 null / 50 signal toys, $N_D=5000$; purity = round-1 pool purity at $f=0.03$; † = the label-free cut had declined at the crossing fraction (the space is then the frozen one at that fraction); purity ‘-’ = declined at $f=0.03$; an ‘(n/m)’ row is an arm still running.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.892	0.05	.029	.052	.078	.037	>.1	.015	.048	.031	.022	.023	.015	.073	.066	.035	.042
SupCon+SIGReg ($\lambda=5$)	0.762	0.13	.023	>.1	>.1	.033	>.1	.015	.029	.029	.040	.037	.014	>.1	>.1	.028	.043
NPLM-dist. (sup.)	0.787	0.13	>.1	>.1	>.1	.021	>.1	.019	.063	.037	.026	.024	.029	.032	.087	.022	.031
SimCLR	0.829	0.05	.029	.046	>.1	.036	.070	.043	.039	.070	.045	.047	.078	.081	>.1	.095	>.1
SimCLR+SIGReg	0.836	0.06	.048	.065	>.1	.034	>.1	.035	.029	.039	.080	.062	.046	.095	>.1	.046	.093
NPLM-bilinear	0.808	0.08	.078	.042	>.1	.026	>.1	.038	.039	.056	.046	.052	.052	.070	.093	.049	.088
GCD (transductive)	0.899	0.30	.009	.013	.009	.031	.017	.019	.025	.029	.024	.019	.009	.017	.021	.031	.025
GCD+SIGReg (transd.)	0.872	0.48	.015	.019	.019	.043	.025	.014	.035	.040	.042	.029	.013	.027	.026	.030	.028
SupCon $\rightarrow$ res (residual)	0.910	0.11	.085	.048	>.1	.024	.025	.021	.040	.045	.043	.039	.036	.063	.072	.044	.040
SupCon $\rightarrow$ res (concat)	0.945	0.12	.084	.041	>.1	.024	.034	.019	.040	.042	.042	.038	.029	.056	.046	.042	.026
SupCon $\rightarrow$ res-nplm (residual)	0.887	0.08	>.1	.027	>.1	.033	>.1	.016	.041	.073	.045	.032	.029	.072	.089	.040	.078
SupCon $\rightarrow$ res-nplm (concat)	0.931	0.09	.070	.034	.070	.034	.052	.015	.042	.031	.026	.024	.023	.063	.073	.031	.041
SupCon+SIGReg $\rightarrow$ res (residual)	0.854	0.06	>.1	.043	>.1	.019	.041	.019	.036	.045	.040	.031	.043	.089	.081	.037	.072
SupCon+SIGReg $\rightarrow$ res (concat)	0.855	0.09	.031	.063	>.1	.021	.042	.017	.026	.037	.036	.031	.019	.092	.085	.028	.041

Table 16: Galaxy10 / lejepa: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / LeJEPA, 14 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.019	.044	.038	.023	.027	.027
Eucl. (anch.)	.010	.020	.018	.014	.020	.018
Maha.	.021	.026	.045	.015	.070	.026
MMD	.010	.071	.046	.019	.040	.040
SparKer	.010	.047	.044	.028	.026	.028
SparKer (anch.)	.010	.036	.042	.025	.025	.025
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.01, d3: 0.02, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.038	.070	.079	.024	.069	.069
Eucl. (anch.)	.010	.020	.016	.014	.020	.016
Maha.	.018	.047	.072	.023	.056	.047
MMD	.018	.047	.042	.028	.041	.041
SparKer	.017	.062	.084	.019	.043	.043
SparKer (anch.)	.010	.052	.069	.018	.028	.028
<i>NPLM-dist. (sup.)</i> — engaged from d0: never, d3: never, d5: 0.05, d7: 0.1, d8: 0.02; median engaged purity 0.59						
Eucl.	>.1	.070 [†]	>.1	.021 [†]	.029 [†]	.070
Eucl. (anch.)	—	—	—	—	.020	.020
Maha.	>.1	>.1	>.1	.021 [†]	.045	.150
MMD	.016 [†]	.052 [†]	.037	.027 [†]	.027 [†]	.027
SparKer	.029 [†]	.028 [†]	.088 [†]	.023 [†]	.028 [†]	.028
SparKer (anch.)	—	—	—	—	.020	.020

Table 17: Galaxy10 / lejepa: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / lejepa, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 82 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.906	0.10	>.1	.065	>.1	.042	>.1	.019	.031	.034	.029	.019	.031	.047	.065	.021	.030
SupCon+SIGReg ($\lambda=5$)	0.823	0.09	>.1	.065	>.1	.029	>.1	.025	.018	.033	.037	.033	.036	.046	.076	.027	.043
NPLM-dist. (sup.)	0.738	0.13	>.1	.065	>.1	.019	.052	.024	.065	.043	.028	.029	.032	.040	.094	.017	.024
SimCLR	0.847	0.04	.075	.047	>.1	.031	>.1	.049	.044	.066	.052	.049	.072	.093	.090	.082	>.1
SimCLR+SIGReg	0.835	0.05	>.1	.090	>.1	.038	>.1	.040	.026	.044	.070	.046	.069	.091	>.1	.042	.092
NPLM-bilinear	0.672	0.06	>.1	.096	>.1	>.1	>.1	.044	.033	.034	.059	.034	.069	.063	.047	.097	>.1
GCD (transductive)	0.886	0.40	.015	.009	.012	.041	.018	.024	.021	.026	.029	.028	.010	.015	.017	.027	.019
GCD+SIGReg (transd.)	0.852	0.48	.013	.017	.026	.031	.026	.017	.024	.039	.033	.027	.022	.022	.032	.029	.019
SupCon $\rightarrow$ res (residual)	0.930	0.06	>.1	.037	>.1	.043	.037	.016	.025	.035	.036	.036	.026	.087	.082	.047	.043
SupCon $\rightarrow$ res (concat)	0.947	0.09	>.1	.040	.084	.031	.042	.019	.026	.034	.026	.025	.025	.045	.063	.028	.029
SupCon $\rightarrow$ res-nplm (residual)	0.890	0.11	>.1	.052	>.1	.024	.073	.019	.036	.033	.047	.027	.031	.065	.081	.022	.029
SupCon $\rightarrow$ res-nplm (concat)	0.923	0.12	>.1	.048	>.1	.033	.075	.018	.035	.033	.029	.024	.029	.049	.071	.023	.029
SupCon+SIGReg $\rightarrow$ res (residual)	0.867	0.05	>.1	.056	>.1	.017	.052	.025	.024	.039	.045	.038	.038	.086	.080	.043	.062
SupCon+SIGReg $\rightarrow$ res (concat)	0.888	0.09	>.1	.042	>.1	.019	.036	.021	.021	.033	.040	.029	.029	.043	.052	.030	.037

Table 18: Galaxy10 / visreg: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / VISREG, 14 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.02, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.070	.038	.063	.024	.028	.038
Eucl. (anch.)	.015	.020	.021	.019	.020	.020
Maha.	>.1	.024	.056	.021	.042	.042
MMD	.017	.037	.040	.029	.027	.029
SparKer	.025	.056	.063	.024	.033	.033
SparKer (anch.)	.022	.042	.044	.021	.021	.022
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.01, d3: 0.01, d5: 0.01, d7: 0.01, d8: 0.02; median engaged purity 1.00						
Eucl.	.077	.048	>.1	.016	>.1	.077
Eucl. (anch.)	.017	.022	.016	.013	.020	.017
Maha.	>.1	.026	.097	.023	.059	.059
MMD	.019	.036	.039	.041	.034	.036
SparKer	.034	.030	.062	.026	.045	.034
SparKer (anch.)	.018	.029	.040	.017	.025	.025
<i>NPLM-dist. (sup.)</i> — engaged from d0: 0.02, d3: 0.05, d5: 0.1, d7: 0.03, d8: 0.1; median engaged purity 0.43						
Eucl.	.082 [†]	.041	>.1	.015 [†]	.026 [†]	.041
Eucl. (anch.)	—	—	—	.030	—	.030
Maha.	>.1	.039	>.1	.016 [†]	.044 [†]	.044
MMD	.018	.044	.044 [†]	.025	.037 [†]	.037
SparKer	.034 [†]	.032	.087	.017 [†]	.027 [†]	.032
SparKer (anch.)	—	—	—	.030	—	.030

Table 19: Galaxy10 / visreg: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / visreg, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 82 (test, draw) points.

space	probe	per-ev.	Maha.					MMD					SparKer				
			c2	c6	c5	c4	c3	c2	c6	c5	c4	c3	c2	c6	c5	c4	c3
SupCon	0.879	0.06	>.1	>.1	.063	.076	>.1	.031	.029	.025	.042	.037	.078	>.1	>.1	.049	.083
SupCon+SIGReg ($\lambda=5$)	0.804	0.04	>.1	.072	.069	>.1	>.1	.021	.025	.031	.047	.029	>.1	>.1	>.1	.079	>.1
NPLM-dist. (sup.)	0.775	0.10	>.1	>.1	>.1	.059	>.1	.037	.023	.029	.038	.037	.054	.086	.070	.038	.048
SimCLR	0.790	0.05	>.1	.066	>.1	>.1	>.1	.029	.039	.056	.067	.052	.070	.097	>.1	.066	>.1
SimCLR+SIGReg	0.801	0.04	>.1	.069	>.1	.076	>.1	.027	.029	.041	.088	.059	.069	.084	>.1	.059	.091
NPLM-bilinear	0.684	0.04	>.1	>.1	.082	>.1	>.1	.097	.042	.041	.093	>.1	.081	.089	.079	.083	.089
GCD (transductive)	0.821	0.07	>.1	.052	.037	.076	>.1	.019	.025	.031	.040	.028	.059	.087	.085	.069	.092
GCD+SIGReg (transd.)	0.831	0.12	.084	.066	.056	.073	>.1	.024	.026	.034	.039	.034	.052	>.1	.096	.066	.098
SupCon $\rightarrow$ res (residual)	0.891	0.05	>.1	.080	.062	.097	>.1	.023	.029	.047	.056	.033	.090	.098	.097	.081	.097
SupCon $\rightarrow$ res (concat)	0.903	0.06	>.1	>.1	.063	.074	>.1	.025	.025	.029	.048	.034	.074	>.1	.098	.073	.092
SupCon $\rightarrow$ res-nplm (residual)	0.862	0.07	>.1	>.1	.056	.087	>.1	.024	.024	.036	.039	.032	.077	>.1	>.1	.047	.092
SupCon $\rightarrow$ res-nplm (concat)	0.890	0.06	>.1	>.1	.063	.075	>.1	.029	.026	.027	.042	.033	.048	.093	>.1	.047	.092
SupCon+SIGReg $\rightarrow$ res (residual)	0.852	0.06	>.1	.073	.092	>.1	>.1	.025	.025	.044	.049	.032	.088	>.1	.096	.082	>.1
SupCon+SIGReg $\rightarrow$ res (concat)	0.869	0.05	>.1	.079	.072	.096	>.1	.021	.025	.036	.047	.033	.089	>.1	>.1	.075	>.1

Table 20: Galaxy10 / dino: pre-discovery $f^*(2\sigma)$, every (space, draw) point. ‘>.1’ = the test never reached 2σ by $f=0.1$ on that draw. *Coverage:* Galaxy10 / DINO, 14 spaces $\times$ 5 single-holdout draws (columns named by the held-out class); exp-150 battery, 200 null / 50 signal toys, $N_D=5000$, fractions 0.006–0.1. GCD arms trained on the unlabelled corpus with the novel images present (transductive).

test	d0	d3	d5	d7	d8	median
<i>SupCon</i> — engaged from d0: 0.01, d3: 0.02, d5: 0.03, d7: 0.01, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	.090	.022	>.1	>.1	.150
Eucl. (anch.)	.019	.021	.030	.014	.018	.019
Maha.	>.1	>.1	.066	>.1	>.1	.150
MMD	.029	.039	.036	.059	.044	.039
SparKer	.062	.079	>.1	.048	.079	.079
SparKer (anch.)	.044	.044	.089	.035	.042	.044
<i>SupCon+SIGReg</i> ($\lambda=5$) — engaged from d0: 0.02, d3: 0.01, d5: 0.01, d7: never, d8: 0.01; median engaged purity 1.00						
Eucl.	>.1	>.1	.062	>.1	>.1	.150
Eucl. (anch.)	.020	.019	.022	—	.031	.021
Maha.	>.1	.075	.078	>.1	>.1	.150
MMD	.027	.029	.039	.046 [†]	.046	.039
SparKer	.048	.077	.084	.075 [†]	>.1	.077
SparKer (anch.)	.044	.046	.047	—	.052	.047
<i>NPLM-dist. (sup.)</i> — engaged from d0: never, d3: never, d5: never, d7: never, d8: 0.03; median engaged purity 0.28						
Eucl.	.071 [†]	>.1	>.1	.018 [†]	>.1	.150
Eucl. (anch.)	—	—	—	—	.030	.030
Maha.	>.1	>.1	>.1	.072 [†]	>.1	.150
MMD	.035 [†]	.024 [†]	.029 [†]	.037 [†]	.052	.035
SparKer	.045 [†]	.078 [†]	.056 [†]	.045 [†]	.094	.056
SparKer (anch.)	—	—	—	—	.047	.047

Table 21: Galaxy10 / dino: post-discovery $f^*(2\sigma)$, every (arm, test, draw) point. [†] = the derived cut had declined at the crossing fraction (the space there is the frozen one); ‘>.1’ = never reached 2σ . *Eucl. (anch.)* and *SparKer (anch.)* consume the discovered anchors. *Coverage:* Galaxy10 / dino, discovery mode: density-ratio pool, derived cut, $n_{\min}=5$, skip-on-refuse; head-only loop (2 rounds $\times$ 5 epochs) over the cached trunk banks; injection from the train bank (clamped at the holdout pool size); 200 null / 50 signal toys; 80 (test, draw) points.